

CLONES, Ireland

WMO: 039740

Lat: 54.183N

Lon: 7.233W

Elev: 292

StdP: 14.54

Time Zone: 1.00 (IRL)

Period: 98-08

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.3	29.7	24.1	18.4	32.3	26.4	20.6	33.8	29.3	46.9	26.9	47.2	4.6	50	0.620

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	72.7	63.5	69.7	61.6	67.2	60.2	64.9	70.2	63.1	67.8	61.6	65.7	8.1	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	86.7	67.6	61.2	81.5	65.4	59.7	77.3	63.7	30.0	70.6	28.7	67.8	27.6	65.8	71.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.0	21.0	18.6	21.7	78.8	2.7	3.2	19.8	81.1	18.2	83.0	16.7	84.8	14.8	87.1		
			21.3	68.5	2.5	1.9	19.5	69.9	18.0	71.0	16.6	72.0	14.7	73.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.5	41.5	41.8	43.8	46.9	52.0	56.4	59.2	59.1	56.1	50.3	45.2	41.5
	DBStd	8.01	4.94	4.91	4.70	4.71	4.04	4.23	4.02	3.48	4.05	4.66	4.46	5.30
	HDD50	1311	264	231	200	113	24	3	0	0	4	55	152	265
	HDD65	5660	728	648	658	542	404	259	188	187	268	456	593	729
	CDD50	1141	1	3	7	22	85	195	285	282	187	64	9	1
	CDD65	12	0	0	0	0	0	1	7	4	0	0	0	0
	CDH74	64	0	0	0	0	0	6	48	10	0	0	0	0
	CDH80	4	0	0	0	0	0	0	4	0	0	0	0	0
Wind	WSAvg	9.2	10.5	10.2	9.9	9.3	8.6	8.3	7.5	7.6	8.6	9.6	10.0	10.1
Precipitation	PrecAvg	37.1	3.5	2.8	3.0	2.4	2.5	2.7	2.6	3.5	3.2	3.8	3.5	3.6
	PrecMax	45.9	6.4	6.5	5.4	4.2	5.7	5.5	5.4	6.1	6.2	7.5	5.6	7.0
	PrecMin	25.8	0.9	0.0	1.0	0.7	0.6	1.0	0.7	0.4	0.2	1.5	1.6	0.9
	PrecStd	4.4	1.4	1.5	1.1	0.9	1.2	1.3	1.1	1.4	1.4	1.6	1.2	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	55.8	57.6	65.1	71.0	74.7	79.7	74.8	70.7	63.1	58.0	55.3
		MCWB	51.6	53.2	52.2	55.1	59.5	63.6	68.1	65.9	62.6	58.3	54.7	52.9
	2%	DB	52.5	53.3	55.2	60.4	66.5	70.4	74.0	70.4	67.4	60.6	55.5	53.0
		MCWB	50.4	50.6	50.9	52.0	57.7	61.2	64.7	63.0	61.4	56.9	53.4	51.5
	5%	DB	50.8	51.3	53.2	57.4	62.9	67.1	69.8	67.9	64.8	58.9	53.8	51.2
		MCWB	48.9	48.9	49.3	51.0	55.3	59.4	61.9	61.4	59.8	55.9	51.8	49.5
	10%	DB	49.1	49.5	51.2	54.9	60.0	64.4	66.5	65.7	62.9	57.3	52.4	49.4
		MCWB	47.3	47.1	48.1	49.4	53.4	57.6	60.0	60.0	58.5	54.4	50.4	47.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.5	53.3	54.3	56.3	61.1	64.7	68.7	67.6	64.8	59.6	55.6	53.5
		MCDB	54.0	55.0	55.9	61.0	68.5	71.4	78.1	72.8	67.8	61.3	56.7	54.4
	2%	WB	50.8	51.2	52.1	54.1	58.5	62.2	65.2	64.7	62.8	58.0	53.9	51.6
		MCDB	52.3	52.7	54.1	58.6	65.2	68.9	72.3	68.4	66.0	59.9	55.1	52.9
	5%	WB	49.2	49.3	50.3	51.9	56.4	60.1	62.9	62.9	61.1	56.3	52.3	49.8
		MCDB	50.6	51.0	52.2	56.1	61.3	65.7	68.5	66.2	64.0	58.3	53.6	51.1
	10%	WB	47.5	47.5	48.5	50.1	54.5	58.5	61.0	61.3	59.3	54.8	50.7	47.9
		MCDB	48.9	49.3	50.7	54.1	58.7	62.9	65.3	64.2	62.0	56.8	52.2	49.2
Mean Daily Temperature Range	5% DB	MDBR	8.7	10.2	11.8	13.5	14.3	13.2	13.2	12.9	12.2	10.7	9.3	8.9
		MCDBR	9.8	10.4	13.3	16.6	18.9	17.3	18.5	15.7	14.2	11.5	9.8	10.9
		MCWBR	9.3	9.1	9.3	9.9	11.3	9.8	10.0	9.1	8.8	8.4	9.0	10.3
	5% WB	MCDBR	9.9	9.6	10.8	14.4	16.5	15.8	16.8	13.2	12.8	10.3	9.3	10.9
		MCWBR	9.7	9.1	8.9	9.5	10.6	9.6	10.1	9.2	9.2	8.4	9.0	10.6
Clear-Sky Solar Irradiance	taub	0.335	0.343	0.367	0.378	0.385	0.375	0.377	0.361	0.354	0.342	0.327	0.324	
	taud	2.225	2.304	2.271	2.274	2.293	2.352	2.353	2.411	2.428	2.403	2.288	2.191	
	Ebn at Noon	187	234	254	269	273	277	274	272	259	233	190	162	
	Edh at Noon	20	26	33	37	39	37	36	32	28	23	19	17	
All-Sky Solar Radiation	RadAvg	199	416	744	1175	1485	1521	1384	1141	844	492	247	150	
	RadStd	20	36	77	123	129	158	123	100	43	43	27	16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page